

Is Korean’s Higher LLM Token Cost Information or Encoding?

A pre-registered code-length measurement on an aligned Korean–English corpus

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Abstract

The same message costs more tokens in Korean than in English under present-day large-language-model (LLM) tokenizers. Two explanations compete: the Korean text may carry more information (an *information* hypothesis), or an English-centric tokenizer may encode Korean inefficiently (an *encoding* hypothesis). Token counts alone cannot separate them. Following Shannon, we treat a tokenizer as a code and a fixed language model as a codebook, and measure each message’s model-relative code length $B_M(s) = -\sum_j \log_2 P_M(z_j \mid z_{<j})$ in bits. On 7,046 aligned sentence pairs from 278 Global Voices articles, held out until the analysis plan was cryptographically frozen, the Korean side costs 1.389 times as many `o200k_base` tokens as the English side (95% cluster-bootstrap CI 1.371–1.406) but also 1.312 times as many bits under Qwen3-1.7B-Base (1.296–1.328). The pooled efficiency ratio $R_\eta = R_B/R_T = 0.945$ (0.936–0.953) implies a 5.5% shortfall in bits per Korean token. The pre-registered rule for the primary endpoint required the one-sided 95% upper bound of R_η to fall below 0.95; the observed bound was 0.952, so the confirmatory rule was not met, while the token-ratio and tokenizer-control comparisons passed ($C_K = 0.867$: a Korean-trained tokenizer encodes the same Korean in 13% fewer tokens). Exploratory analyses show the shortfall is larger for short messages (9.3% in the shortest tercile) and that it would have been 41% and 71% under the previous two OpenAI tokenizer generations. We argue that the residual Korean token surplus under a current tokenizer is dominated by model-relative bits rather than by encoding waste, and that the remaining encoding effect, while consistently present, is small.

1 Introduction

Users of LLM application programming interfaces (APIs) are billed per token, and a widely repeated observation is that Korean text is charged for substantially more tokens than English text of the same meaning. The observation is usually attributed to tokenizers trained predominantly on English, which split Korean into many short byte-level fragments. That attribution is plausible, but it is not the only one. Korean and English versions of a message are translations, not copies: the Korean may add honorific marking, explicit topic particles, or explanatory expansion, and a model may find Korean harder to predict. Both mechanisms raise the token count, so the token count itself cannot tell them apart.

The information-theoretic view separates them cleanly. A tokenizer is a code that maps text to symbols; how many symbols a message needs is a property of the code. How much information the message carries is a property of the message and of the receiver’s predictive model. If Korean needs more tokens *and* more bits in proportion, the surplus is information. If Korean needs more tokens but not more bits, the surplus is encoding.

This paper reports a measurement designed to make that distinction with the decision rules fixed in advance. The contributions are: (i) an estimand, the pooled efficiency ratio R_η , that isolates encoding inefficiency from information content up to a stated model-relative caveat; (ii) a

pipeline with unit tests that computes it reproducibly with pinned model, tokenizer, and corpus revisions; (iii) a pilot/held-out split with thresholds frozen by content hash before the held-out data were scored; and (iv) the confirmatory result, which narrowly fails the pre-registered primary rule, together with exploratory analyses that explain why.

2 Background: tokenizers as codes, models as codebooks

Shannon’s source-coding theorem distinguishes the information in a message from the number of symbols used to write it down [7]. An ideal code assigns to a symbol of probability p a codeword of length $-\log_2 p$ bits; a code adapted to a different source wastes symbols. Shannon’s later experiment on printed English estimated the entropy of a language from a human predictor’s guesses [8]; a language model plays the same predictive role today, with the difference that its predictions are explicit probabilities.

Let M be a fixed autoregressive language model with its own tokenizer τ_M . For a message s , write $z = \tau_M(s) = (z_1, \dots, z_n)$. The *model-relative code length* is

$$B_M(s) = - \sum_{j=1}^n \log_2 P_M(z_j \mid z_{<j}), \quad (1)$$

the number of bits an arithmetic coder driven by M would spend to transmit s [2]. Two properties matter. First, $B_M(s)$ does not depend on any *billing* tokenizer: it is computed through τ_M only. Second, it is model-relative. Writing P for the true conditional distribution and Q for the model’s, the expected code length decomposes as $H(P, Q) = H(P) + D_{\text{KL}}(P \parallel Q)$, so B_M includes the model’s own prediction error. A model that knows Korean less well than English will overprice Korean text; the direction of this bias is discussed in Section 5.

For a billing tokenizer b with $T_b(s) = |\tau_b(s)|$, the efficiency of b on s is the number of bits carried per billed token, $\eta_{M,b}(s) = B_M(s)/T_b(s)$. The corpus-level estimands are ratios of sums over aligned pairs i :

$$R_T = \frac{\sum_i T_{i,\text{KR}}}{\sum_i T_{i,\text{EN}}}, \quad R_B = \frac{\sum_i B_{i,\text{KR}}}{\sum_i B_{i,\text{EN}}}, \quad R_\eta = \frac{\sum_i B_{i,\text{KR}} / \sum_i T_{i,\text{KR}}}{\sum_i B_{i,\text{EN}} / \sum_i T_{i,\text{EN}}} = \frac{R_B}{R_T}. \quad (2)$$

Under the information hypothesis $R_B \approx R_T$ and $R_\eta \approx 1$; under the encoding hypothesis $R_T \gg 1$, $R_B \approx 1$, and $R_\eta < 1$. The quantity $1 - R_\eta$ is the relative shortfall in bits per Korean token compared with an English-efficiency counterfactual. It is not a literal count of empty tokens. Ratios of sums are used rather than means of per-message ratios because the latter are biased upward by short messages and do not satisfy the identity $R_\eta = R_B/R_T$.

An independent control uses a tokenizer trained on Korean, k . The Korean text and therefore its information are unchanged; only the code changes. The absolute Korean-token reduction

$$C_K = \frac{\sum_i T_{i,\text{KR}}^{(k)}}{\sum_i T_{i,\text{KR}}^{(b)}} \quad (3)$$

measures how much of the Korean cost under b is removable by a different code. $C_K < 1$ shows tokenizer dependence; attributing it to training-language balance alone would require holding the algorithm, vocabulary size, normalisation, and pre-tokenisation fixed, which two off-the-shelf tokenizers do not.

3 Measurement design

3.1 Corpus and split

We use the OPUS packaging of Global Voices v2018q4 in English and Korean [9], a citizen-media news collection whose articles are translated by volunteers. The archive contains 365 article-level

alignment groups, of which 347 contain aligned text, yielding 9,017 aligned sentence pairs. Article identity is preserved as a cluster identifier; the corpus has one genre (news), and the translation direction is not recorded.

The release has no train/test division, so a deterministic article-level split was created before any scoring. Articles connected by an identical bilingual pair were merged into indivisible components; components were ranked by a seeded SHA-256 rule (seed 20260910) and the prefix closest to 20% of articles was assigned to the pilot. The pilot contains 69 articles (1,732 pairs) and the held-out set 278 articles (7,285 pairs); no article or exact bilingual pair crosses the split. The held-out file was neither inspected nor scored until the specification was frozen.

Only Unicode NFC normalisation and surrounding-whitespace removal are applied. Validation flags record rows whose Korean side contains no Hangul, rows whose Korean and English text are identical (untranslated photo credits and licence strings), disallowed control characters, and length statistics.

3.2 Estimator model and scoring policy

The estimator is Qwen3-1.7B-Base [5] at revision `ea980cb0`, run in `bfloat16` with the log-softmax computed in `float32`. Every message is scored in evaluation and inference mode from raw next-token logits. The model’s document-boundary token (`<|endoftext|>`, id 151643) is prepended and every content token is scored conditional on it, so that the first content token receives a code length; no end-of-sequence event is charged. Special tokens are never added implicitly (`add_special_tokens=False`), padding is never scored, and the pipeline asserts on every call that the number of scored targets equals the number of content tokens. The longest message (587 characters) is far below the 32,768-token context, so no chunking occurred.

3.3 Tokenizers

The baseline billing tokenizer is OpenAI’s `o200k_base` encoding (tiktoken 0.14.0; 199,998 mergeable tokens), used by the GPT-4o generation [4]. The Korean-trained comparison tokenizer is that of EleutherAI’s Polyglot-Ko-1.3B [1] at revision `557e162c`, a 30,000-entry byte-level BPE trained on a Korean-only corpus. Token counts are content-only, with no chat template or wrapper tokens. Counting tokenizers return integers only; their token ids are never fed to the estimator, and the estimator is scored only through its own tokenizer. All tokenizer files are hashed and the hashes recorded in the provenance of every run.

3.4 Pre-registration

Ratios described as “well above one” or “close to one” are meaningless until thresholds are fixed, and thresholds chosen after seeing the data prove nothing. All thresholds, endpoints, and exclusion rules were therefore written into a specification file whose content hash (SHA-256 `850fa9be...`) was recorded when it was frozen on 10 September 2026, after the pilot had been analysed and before the held-out set was scored. The analysis program refuses to run in confirmatory mode on an unfrozen or subsequently edited specification and refuses pilot-tagged input files. The frozen decisions are listed in Table 1.

The pilot informed these choices in two ways. Its point estimates ($R_\eta = 0.940$, $R_T = 1.377$, $C_K = 0.870$ after exclusions) located the effect, and a precision simulation, which resampled the pilot’s per-article sums to the held-out article count, described the distribution of the bootstrap bounds that the rules would use. For $c_\eta = 0.95$ that simulation gave a pass probability of 0.78 conditional on the pilot values carrying over unchanged; the threshold lay 1.25 pilot standard errors above the pilot point estimate. This is recorded here because it bears directly on the interpretation of the confirmatory outcome.

Table 1: Frozen decisions. Bounds are cluster-bootstrap percentiles.

Decision	Value	Rule on the held-out bootstrap
Primary endpoint	R_η	one-sided 95% upper bound $< c_\eta = 0.95$
Token ratio	R_T	one-sided 95% lower bound $> c_T = 1.20$
Tokenizer control	C_K	one-sided 95% upper bound $< c_C = 0.90$
Overall rule	all three confirmatory comparisons must pass	all_confirmatory_pass
Bit-ratio equivalence (exploratory)	R_B	90% CI inside [0.8333, 1.20]
Exclusions	identical KR/EN text; no Hangul on KR side	applied identically to pilot and held-out
Bootstrap	10,000 replicates, seed 20260910	whole articles resampled, both languages together

3.5 Statistical analysis

All estimands are ratios of corpus sums, Eq. (2). Uncertainty is quantified by a paired, stratified cluster bootstrap: articles are resampled with replacement within genre strata, every pair of a resampled article enters with both its Korean and English measurements, and each pooled ratio is recomputed in every replicate. Because the estimands are functions of column sums only, per-article sums are precomputed and the bootstrap is exact. The identity $R_\eta = R_B/R_T$ is asserted in every replicate. One-sided decision bounds are the 5th or 95th percentiles; equivalence for R_B uses the two one-sided tests procedure, requiring the 90% percentile interval to lie inside the pre-specified interval [6].

Exploratory analyses, which never feed the confirmatory rule, are: pooled estimands within terciles of English character count; the distribution of per-message ratios; and paired cluster-level sign-flip permutation tests in which the Korean and English measurements of all pairs in a randomly chosen set of articles are exchanged, with the log pooled ratio as statistic and 10,000 permutations. The three language ratios are functionally dependent and are reported as one finding seen from three angles.

3.6 Implementation

The pipeline is a Python package (**klen**) with 66 unit tests, including hand-verifiable code-length cases (a uniform fake model yields exactly $\log_2 10$ bits per token; a biased fake model exactly 1 bit), whole-cluster resampling verified against the enumerable set of possible replicates, stratum preservation, freeze and tamper detection, and an independent re-implementation cross-check against a tiny random model. Provenance for every run records package versions (PyTorch 2.14.0, Transformers 5.17.0, Python 3.13.1), tokenizer file hashes, the dataset hash, the device (Apple M4 Pro, MPS backend), the specification, and the validation report. The held-out run took 34 minutes.

4 Results

4.1 Sample

After the frozen exclusion rules, the held-out set comprises 7,046 pairs in 278 articles (239 rows dropped: 104 with identical Korean and English text and a further 135 with no Hangul on the

Korean side). The pilot after the same rules comprises 1,672 pairs in 69 articles. No row changed under NFC normalisation; no pair id was duplicated.

4.2 Pooled estimands and confirmatory decisions

Table 2: Held-out pooled estimands (278 articles, 7,046 pairs) with 95% cluster-bootstrap percentile intervals, and the pre-registered decisions. R_T and C_K use o200k_base; C_K and the Korean-fit row use Polyglot-Ko. Bits are Qwen3-1.7B-Base code lengths.

Quantity	Point	95% CI	Decision bound	Result
R_T token ratio KR/EN	1.3888	[1.3714, 1.4058]	lower 1.3741 > 1.20	pass
R_B bit ratio KR/EN	1.3121	[1.2959, 1.3278]	90% CI [1.2984, 1.3252], outside [0.8333, 1.20]	exploratory
$R_\eta = R_B/R_T$ (primary)	0.9448	[0.9363, 0.9532]	upper 0.9519 $\not<$ 0.95	fail
$1 - R_\eta$ shortfall	0.0552	[0.0468, 0.0637]	—	—
C_K Korean tokens, Polyglot/o200k	0.8665	[0.8601, 0.8729]	upper 0.8719 < 0.90	pass
R_T under Polyglot-Ko	0.5095	[0.5025, 0.5163]	—	exploratory
G gap contraction	0.3668	[0.3629, 0.3707]	—	exploratory
Overall (all_confirmatory_pass)				fail

Table 2 gives the held-out estimates. The Korean side costs 38.9% more o200k_base tokens and 31.2% more bits than the English side. The efficiency ratio is $R_\eta = 0.945$: a Korean token carries 5.5% fewer bits than an English token, with a 95% interval of 4.7% to 6.4%. The token-ratio and tokenizer-control comparisons pass their pre-registered rules with wide margins. The primary comparison does not: the one-sided 95% upper bound of R_η is 0.9519, above the frozen threshold of 0.95, and under the overall rule the confirmatory claim is therefore not supported. The bit ratio is far outside the exploratory equivalence interval; Korean text costs more bits as well as more tokens.

The pilot and held-out estimates agree closely (Table 3). The held-out R_η is 0.0048 above the pilot value, about 1.2 held-out standard errors, an ordinary sampling shift; with the threshold placed 1.25 pilot standard errors above the pilot estimate, that shift was enough.

Table 3: Pilot versus held-out point estimates after identical exclusions.

Quantity	Pilot (69 articles, 1,672 pairs)	Held-out (278 articles, 7,046 pairs)
R_T	1.3769	1.3888
R_B	1.2943	1.3121
R_η	0.9400	0.9448
C_K	0.8697	0.8665

4.3 Exploratory analyses

Message length. Table 4 splits the held-out pairs into terciles of English character count. The token surplus, the bit surplus, and the efficiency shortfall all shrink with length: $1 - R_\eta$ is 9.3% in the shortest tercile and 4.4% in the longest, with non-overlapping intervals. Because pooled ratios weight by tokens, the corpus-level estimate is dominated by long messages. Part of the length dependence is mechanical: under the prepend-BOS policy the first content token is scored from a document boundary and is expensive in both languages, and this fixed cost is a larger fraction of a short message.

Table 4: Held-out pooled estimands within terciles of English character count (exploratory). Articles may span terciles; bootstrap within tercile.

Tercile	Chars (EN)	n	R_T	R_B	R_η [95% CI]	C_K
Q1 (short)	3–74	2,349	1.564	1.419	0.907 [0.894, 0.921]	0.882
Q2	74–137	2,348	1.415	1.318	0.931 [0.919, 0.944]	0.868
Q3 (long)	137–587	2,349	1.329	1.271	0.956 [0.946, 0.966]	0.861

Per-message distribution. Per-message ratios are descriptive only. The median per-pair token ratio is 1.429 (5th–95th percentiles 0.762–2.364), the median bit ratio 1.357 (0.766–2.091), and the median efficiency ratio 0.944 (0.666–1.369). Their arithmetic means (1.489, 1.396, 0.973) exceed the pooled estimates, illustrating the upward bias of averaging ratios.

Permutation tests. Exchanging the Korean and English measurements of whole articles 10,000 times gives observed log pooled ratios of +0.328 (R_T), +0.272 (R_B), -0.057 (R_η) and -0.143 (C_K), each with two-sided $p = 1/10,001$, the smallest attainable value. The sign and existence of every effect are not in doubt; the confirmatory question was only its size.

4.4 Post hoc: the baseline tokenizer generation

To understand why the effect is small, we recomputed R_T and R_η on the held-out set holding the Qwen3 bits fixed and substituting the two previous OpenAI encodings for the billing tokenizer (Table 5). This was done after the confirmatory analysis and is not part of it.

Table 5: Held-out R_T and R_η under three OpenAI encodings with the same Qwen3-1.7B-Base bits (post hoc, exploratory).

Encoding	Model generation	R_T	R_η
r50k_base	GPT-3	4.591	0.286
c1100k_base	GPT-3.5 / GPT-4	2.238	0.586
o200k_base	GPT-4o (frozen)	1.389	0.945

Under the GPT-3 encoding a Korean token carried 71% fewer bits than an English token; under the GPT-4 encoding 41% fewer; under the current encoding 5.5% fewer. Most of the historical Korean “token tax” has been removed by vocabulary growth and multilingual training of the tokenizer itself. The residual surplus under o200k_base is dominated by the bit ratio.

5 Discussion

5.1 Why the primary comparison failed

Two distinct things happened. Statistically, the threshold left almost no room: it was set 1.25 pilot standard errors above the pilot estimate, the pre-freeze simulation gave the rule only a 78% chance of passing even if the pilot value were exactly right, and the held-out estimate moved up by an unremarkable 1.2 standard errors. A threshold of 0.96 would have passed in every simulated sample. None of this justifies revising the threshold now; it is precisely what a pre-registered rule is for.

Substantively, R_η is close to one because R_B is large, not because R_T is small. Under the current tokenizer the Korean token surplus is 39%, and the model spends 31% more bits on the Korean side. Three contributors, which the present design cannot separate, push R_B upward:

1. *Estimator tokenization.* Bits are summed over the estimator’s own tokens, and Qwen3’s tokenizer splits the Korean side into 1.554 times as many tokens as the English side. Finer

segmentation generally raises total code length, so part of the Korean bit surplus is a property of τ_M rather than of the message.

2. *Model fluency.* The D_{KL} term is larger for the language the model predicts less well. A 1.7B-parameter model trained predominantly on English and Chinese plausibly overprices Korean. The held-out code lengths are 4.87 bits per Qwen token for Korean versus 5.77 for English, and 1.42 versus 1.23 bits per UTF-8 byte.
3. *Translation content.* Global Voices Korean articles are volunteer translations that may add explanation; parallel meaning does not guarantee equal information.

The first two contributors bias R_B upward and R_η toward one, that is, *against* the encoding hypothesis. The measured 5.5% shortfall is therefore a lower bound on the encoding effect only in the direction of these biases, and the third contributor could act either way.

5.2 What the evidence supports

The pre-registered claim that Korean tokens carry substantially (at least 5%) less information than English tokens under a present-generation tokenizer was not established at the 95% level. What the data do establish, with narrow intervals and stable pilot-to-held-out agreement, is: (i) a consistent Korean token surplus of about 39%; (ii) a consistent and smaller efficiency shortfall of about 5%, larger for short messages; (iii) that a Korean-trained tokenizer removes about 13% of the Korean tokens without changing the text; and (iv) that under earlier tokenizer generations the encoding shortfall was large. The picture is one in which encoding inefficiency was the dominant cause of Korean’s token cost under older tokenizers and has become a minor cause under the current one, with the remaining surplus tracking model-relative bits.

5.3 Limitations

The estimator is a single small model; bits are Qwen3-1.7B-relative and the fluency bias is unquantified. A replication with a second estimator, and on the multi-genre FLORES+ corpus [3] for which a specification file is prepared, is planned as exploratory work. The corpus is one genre with unknown translation direction, and it predates the estimator’s training cut-off, so training-data contamination is unknown rather than excluded. The three language ratios are functionally dependent. C_K shows tokenizer dependence but not its cause. Finally, the study was run once as a complete test of the pipeline; if the design is revised, a new specification must be frozen and the held-out set re-scored, and the present result would then be reported as the outcome of the first pre-registration.

6 Conclusion

Treating tokenizers as codes and a fixed language model as a codebook turns the Korean token-cost question into a measurable comparison of symbols against bits. On 7,046 held-out aligned pairs the Korean side costs 39% more tokens under `o200k_base` and 31% more Qwen3 bits, an efficiency shortfall of 5.5% (95% CI 4.7–6.4%). The pre-registered rule, which required at least a 5% shortfall with 95% confidence, was narrowly not met. A Korean-trained tokenizer nonetheless encodes the same text in 13% fewer tokens, and under previous tokenizer generations the shortfall would have been 41% and 71%. The encoding tax on Korean is real, has shrunk dramatically with tokenizer vocabulary growth, and under a current tokenizer is now small relative to the model-relative information cost.

Reproducibility

Code, tests, the frozen specification, and all analysis outputs (bootstrap replicates, provenance, and per-pair measurements with corpus text removed) are at <https://github.com/damifiance/>

[korean-token-tax](#). The corpus is rebuilt from the OPUS archive by a script that reproduces the split; model and tokenizer files are fetched at the pinned revisions and hash-verified before every run.

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A Pilot precision simulation

Before the thresholds were chosen, the pilot’s 69 per-article sums were resampled with replacement to 278 articles 400 times; within each simulated sample a 2,000-replicate cluster bootstrap produced the bounds that the confirmatory rules use. Table 6 reports the quantiles of those bounds across simulations. The 95th percentile of the R_η upper bound, 0.953, coincides with the value later observed.

Table 6: Quantiles across 400 simulated held-out samples (278 articles) of the decision bounds, using pilot per-article sums as the population.

Bound	5%	25%	50%	75%	95%
R_T lower 95%	1.3325	1.3443	1.3518	1.3586	1.3684
R_B 90% CI low	1.2563	1.2654	1.2723	1.2782	1.2865
R_B 90% CI high	1.2863	1.2947	1.3013	1.3067	1.3157
R_η upper 95%	0.9399	0.9440	0.9467	0.9495	0.9530
C_K upper 95%	0.8853	0.8902	0.8933	0.8970	0.9030

B Exclusion sensitivity on the pilot

Before the exclusion rules were chosen, every candidate rule was applied to the pilot. Dropping identical-text rows, rows without Hangul, rows shorter than five characters, rows with a Korean/English character ratio outside $[0.2, 1.5]$, or all of these together moved R_η within 0.938–0.941 and R_T within 1.368–1.393. The pooled estimands are insensitive to these choices; the rules adopted remove untranslated material on grounds of validity rather than effect.